

Metaheuristic Algorithms and Applications: Program Report II-1

Program Report on Control Design

Deadline: May 25, 2026

Consider the ball and beam system shown in Figure 1. The beam is made to rotate in a vertical plane by applying a torque at the center of rotation and the ball is free to roll (with one degree of freedom) along the beam. We require that the ball remains in contact with the beam and that the rolling occur without slipping, which imposes a constraint on the rotational acceleration of the beam. The system is described as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} x_2 \\ B(x_1 x_4^2 - G \sin x_3) \\ x_4 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u, \quad y = x_1$$

where $[x_1 \ x_2 \ x_3 \ x_4]^T = [r \ \dot{r} \ \theta \ \dot{\theta}]^T$ and $u = \tau$. The problem is to design a GA, PSO, DNA, IM, GSA, ABC, ..., or Queen-Bee based PID controller, Sliding-mode control, or fuzzy logic controller such that $y(t) = y_d(t) = A \cos \pi t / 5$, with $A=1, 2$, and 3 and initial conditions for (r, θ) of $(1, 0.0564)$, $(2, 0.1129)$, and $(3, 0.1698)$, respectively. The constraints are (a) $-4m \leq x_1 \leq 4m$; (b) $0.6 \leq B \leq 0.8$; (c) $G = 9.81$; and (d) $|u| \leq 20$.

This report should include “programs” and “discussions”. For discussion part, the following questions should be answered at least.

- (i) Which kind of heuristic optimization algorithm or AI method is adopted? Why?
- (ii) Which kind of controller (PID, SMC, FLC, or DNA based PID) does your program utilize? Why?
- (iii) Please explain why the fitness function, all the parameters and settings that are chosen or involved in the design of such controller.

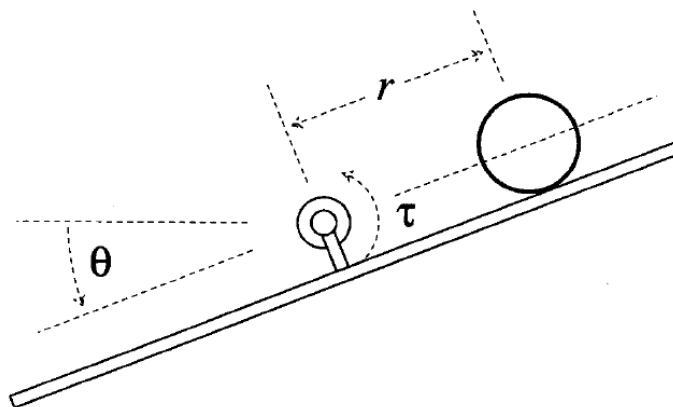


Fig. 1. The ball and beam system.